

**INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
MID-SPRING SEMESTER EXAMINATION 2023-24**

Subject No. : CS60075

Subject : Natural Language Processing

Duration: 2 Hours

Full Marks: 60

Department: Computer Science and Engineering

Specific charts, graph paper, log book etc., required: NA

Special Instructions (if any) :

- *Attempt all questions. All parts of the same question must be answered together.*
 - *Answers should be brief and to-the-point. Sketchy answers will be penalized.*
 - *Make reasonable assumptions only if necessary, and state any assumptions made. No clarification can be provided in the examination hall.*
 - *All working steps must be shown. Writing the answer without showing steps will not be given any credit, even if the answer is correct. You can use calculators.*
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Question 1 (Miscellaneous) [(2 x 3) + 2 + 4 = 12]

(a) State whether the following statements are True or False. Give a short but precise explanation in either case. Marks will not be awarded unless the explanation is correct.

i) Given a well-tuned unigram language model $p(w|\theta)$ trained on all the textbooks in the domain of "Programming and Data Structures", we can conclude that $p(\text{"Programming and Data Structures"}|\theta) > p(\text{"and Structures Programming Data"}|\theta)$.

ii) In general, lemmatization is slower than stemming.

iii) In a document, the number of word types is greater than or equal to the number of word tokens.

(b) In a text corpus, if the frequency and frequency-based rank of 'word1' is 200 and 100 respectively, and the frequency of 'word2' is 800, find the rank of 'word2' according to Zipf's Law. If on average 'word1' has 2 meanings, then how many meanings is 'word2' likely to have?

(c) Mention two problems that can occur in practice if we use the chain rule of probabilities method to find the probability of a sentence. For each problem, suggest a better approach to solve it.

Question 2 (Spelling Correction) [10]

Find the edit distance between the two strings "Pollution" and "Population". Assume that costs of edit operations are as follows: insertion = 1, deletion = 1, substitution = 1. You should show all the steps in your calculations and the edit distance table.

Question 3 (Language Models) [6 + 3 + 6 + 3 + 2 = 20]

Consider the following pre-processed corpus (with stopwords removed) C :

<s> Large Language Models NLP </s>
 <s> Language Models Text Classification </s>
 <s> Approaches Text Generation Classification </s>

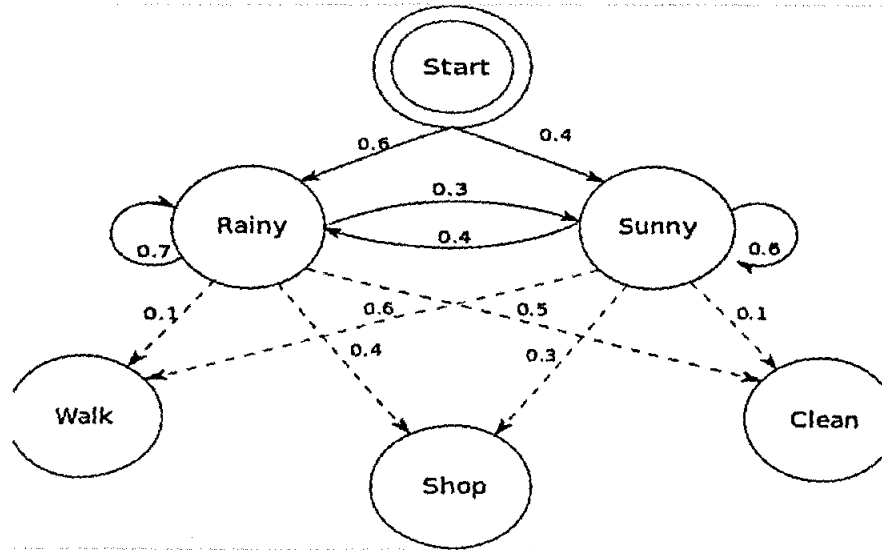
and the pre-processed sentence S : <s> NLP Text Classification Text Generation </s>

- Calculate all unigram probabilities for C . Note that 'NLP' is a single token. You also need to calculate the probabilities for <s> and </s>.
- Calculate the perplexity of the unigram model for sentence S .
- Calculate all bigram probabilities for C . Note that </s><s> is not a valid bi-gram.
- Calculate the perplexity of the bigram model for sentence S , using **Unigram Prior Smoothing (UPS)** which we refer to as P_{bi-ups} . Assume that $m = |V|$, where V is the unigram vocabulary consisting of <s> and </s> as well.

$$P_{bi-ups}(w_i|w_{i-1}) = \frac{c(w_{i-1}w_i) + mP(w_i)}{c(w_{i-1}) + m}$$

- Consider for a corpus, we can calculate the unigram probability $P_{uni}(w_i)$, the bigram probability $P_{bi}(w_i|w_{i-1})$ and $P_{bi-ups}(w_i|w_{i-1})$. What happens to $P_{bi-ups}(w_i|w_{i-1})$ when $m \rightarrow 0$ and when $m \rightarrow \infty$ respectively? [Hint: Answer in terms of $P_{uni}(w_i)$ and $P_{bi}(w_i|w_{i-1})$.]

Question 4 (HMM) [1 + 2 + 2 + 3 = 8]



Rahul will decide his day's work (walking down the street, shopping in a mall or cleaning his room) depending on the day's weather. The probabilities of weather and his mood to do a job are given in the HMM model diagram. Now derive the following probabilities according to the given figure. Show steps for each computation.

- (i) What is the probability that he will shop today when the weather is Sunny?
- (ii) What is the probability that he will clean his room today whatever be the weather?
- (iii) What is the probability that tomorrow will be Sunny?
- (iv) What is the probability that the 4th day will be Rainy given that tomorrow is Sunny (i.e., $t_4 = R \mid t_2 = S$)?

Question 5 (Viterbi) [10]

Find the POS tag of each word of the following sentence using the Viterbi algorithm. Show all computations. Not showing all computations, or wrong computations will not be awarded any marks even if the final POS tags are correct.

No man is an island.

For simplicity, assume that all tags have the same probability 1.0 to appear in the beginning of a sentence.

Emission probabilities:

	No	man	is	an	island
Det	0.78	0	0.007	0.91	0
VB	0.0039	0.058	0.5	0	0.0037
NN	0	0.48	0	0.0083	0.99

Transition Probabilities:

	Det	VB	NN
Start	0.65	0.0086	0.097
Det	0	0.00057	0.64
VB	0.54	0.078	0.09
NN	0.072	0.35	0